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Pattern of Muscle Fiber Type Formation in the Pig

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ABSTRACT **The aim of this study was to analyze the temporal sequence of expression of the myosin isoforms in the populations of muscle fibers in the pig and to bring more information on the origin of the strikingly different pattern of fiber composition and distribution between the deep medial red (oxido-glycolytic) and superficial white (glycolytic) portions of semitendinosus (ST) muscle. Muscle samples were taken from 49-, 55-, 75-, 90-, 103-, and 113- (birth) day-old fetuses, from 6-, 11-, 21-, 35-, 50-, and 80-day-old piglets, and from a 3-year-old pig. Our results confirm the sequential formation of primary and secondary generation fibers. The use of immunohistochemistry and heterologous monoclonal antibodies (mAb) directed against specific myosin heavy chain (MHC) isoforms revealed a different pattern of gene expression between the two portions of the ST muscle for both generations of fibers. By 75 days of gestation (dg), primary myotubes from the deep medial portion stained positively for the anti-slow MHC mAb and negatively for the adult anti-fast MHC, whereas the opposite was observed in the superficial portion. Secondary fibers never expressed slow MHC until late gestation. Instead, they expressed an adult fast MHC isoform as soon as they formed in the deep medial portion and later on in the superficial portion. From late gestation to the first 3 postnatal weeks, slow MHC began to be expressed in a subpopulation of secondary fibers. These fibers were in the direct vicinity of primary myotubes in the deep medial portion, whereas their location could not be established in the superficial portion. The remaining secondary fibers matured to type IIA in the direct vicinity of these type I fibers and to type IIB at the periphery of the islets. In both portions of the muscle, a subpopulation of secondary fibers, the first ones to express slow MHC, also transitorily expressed a MHC that was identical or closely related to the α-cardiac MHC during the early postnatal period. A third generation of small diameter fibers was observed shortly after birth and reacted with the anti-fetal MHC mAb; their destiny remains to be established. The present work reveals a remarkable pattern of MHC gene expression in the pig and raises many questions on the real nature of these isoforms. In order to answer these questions, we have undertaken to make a cDNA library of pig skeletal muscle and to screen this library with the same mAbs used in the present study.**

Key words: Skeletal muscle, Myofiber, Myosin, Development, Pig

INTRODUCTION

Most muscles of the pig exhibit a highly organized pattern and unique distribution of fiber types consisting of islets of slow-twitch (type I) fibers surrounded by an internal rosette of type IIA fibers and an external ring of type IIB fibers (Lefaucheur and Vigneron, 1986). Several studies have been carried out to describe the ontogenesis of this fiber type distribution (Davies, 1972; Thurley, 1972; Ashmore et al., 1973; Swatland and Cassens, 1973; Beermann et al., 1978; Wigmore and Stickland, 1983; Handel and Stickland, 1987). Pig myogenesis as in many mammalian species is a biphasic phenomenon with the sequential formation of two generations of muscle fibers. A primary generation forms from 35 until 55 days of gestation (dg), followed by a second generation which forms between 55 and 90–95 dg. These secondary fibers form around the primary myotubes, using them as a scaffold. The total number of muscle fibers is generally considered to be definitively established by 90–95 dg in the pig and birth occurs at 113 dg. From late gestation through the first postnatal weeks, these fibers undergo a process of maturation when the highly organized pattern encountered in the adult pig is established (Suzuki and Cassens, 1980; Lefaucheur and Vigneron, 1986).

Most of these previous studies used conventional histochemical techniques based on the sensitivity of actomyosin ATPase to acidic and alkaline pH (Brooke and Kaiser, 1970). Although these techniques are well suited to resolve most adult fiber types (I, IIA, IIB) in mature muscle, they do not allow a clear classification of fiber types during the fetal and perinatal periods. During this time, the actomyosin ATPase is generally more or less resistant to both acid and alkaline pH and many fibers behave in the same manner as fibers containing a mixture of adult fast and slow myosins, and as a consequence have often been classified as type IIC fibers. However, these fibers have been shown to contain different myosin heavy chain (MHC) isoforms such

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as embryonic and fetal isoforms (for a review see Pette and Staron, 1990). This makes it difficult to interpret histochemical results in terms of myosin polymorphism. So far, four major MHC isoforms have been identified in adult mammalian limb muscle (I, IIA, IIB, and IIX) (Schiaffino et al., 1989), and two isoforms in developing limb muscles (embryonic and fetal or neonatal) (Whalen et al., 1981). Recently, the use of monoclonal antibodies raised against specific MHC isoforms has led to a more detailed description of fiber specialisation than previously obtained by conventional histochemistry.

In light of these findings, we have re-examined the ontogenesis of the highly organized pattern of fiber type distribution in the pig by combining conventional histochemical and biochemical techniques with immunocytochemistry using heterologous monoclonal antibodies that recognized specific MHC isoforms. The presence of red and white portions within the pig semitendinosus muscle makes it an appropriate model for investigating the phenomenon according to muscle type.

RESULTS

The mature pig semitendinosus muscle contains two portions which exhibit striking differences in fiber composition (Fig. 1). The superficial white (glycolytic) area is mainly composed of fast-twitch (type II) fibers while the deep medial red (oxido-glycolytic) area exibits an extensive grouping of slow-twitch (type I) fibers. In the adult, the percentages of type I, IIA, and IIB fibers were 4, 8, and 88% in the superficial portion and 39, 26, and 35% in the deep medial portion, respectively.

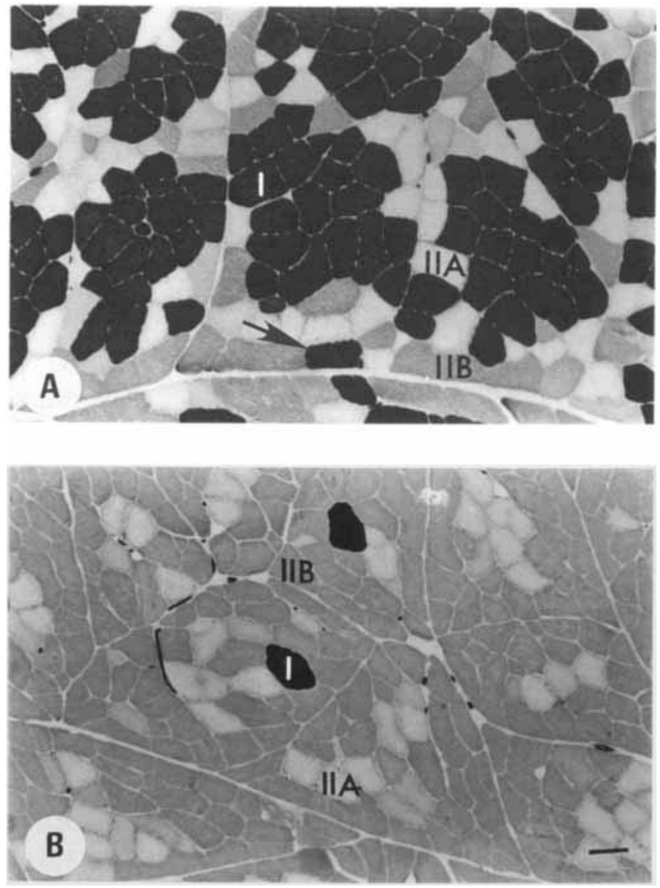


Fig. 1. ATPase activity of the deep medial (**A**) and superficial (**B**) portions of adult pig semitendinosus muscle after pre-incubation at pH 4.4. Scale bar = 100 μm.

Native Myosin Isoforms

For practical reasons, the pattern of myosin isozyme transitions during development was studied in the whole semitendinosus (ST) muscle whithout any distinction between the deep medial and superficial portions. These results are shown in Figure 2. An embryonic (e) isoform with an intermediate mobility between adult fast and slow isoforms was observed at 49 and 55 dg and disappeared between 55 and 75 dg (Fig. 2, lanes a,b,c). Three fetal isoforms (f1, f2, f3) appeared between 55 and 75 dg, concomitantly with the formation of secondary fibers (Fig. 3f,j,h,l). They were present throughout fetal development (Fig. 2, lanes c,d), decreased progressively during late gestation (Fig. 2, lane e), and disappeared around birth (Fig. 2, lane f), in agreement with the immunocytochemical results which showed a decreased reactivity to the anti-fetal antibody at that time (Fig. 4f,j,n, and h,l,p). Adult fast isoforms (F) appeared during the first three post-natal weeks (Fig. 2, lanes g,h,i), concomitantly with the maturation of the types IIA and IIB fibers as demonstrated by conventional histochemistry (Fig. 5). They most likely correspond to the adult isoforms FM1, FM2, and FM3.

Slow myosin isoforms were present throughout development. Their proportion increased soon after birth (Fig. 2, lanes f,g), concomitantly with the maturation of the second generation of slow fibers (Fig. 4i,m). Three slow isoforms (SM1, SM2, and SM3) could be clearly detected from birth onwards (Fig. 2, lanes f,g,h,i) and were strongly expressed in the deep medial portion of adult muscle (Fig. 2, lane j).

The transitory expression of an isoform with an electrophoretic mobility intermediate between the adult fast and slow native isoforms was observed during the weeks following birth (Fig. 2, lanes g,h,i). Comigration experiments with extracts from adult pig atria and ventricle showed that this band had the same mobility as α-cardiac V_1 myosin (Fig. 6).

Electrophoretic Analysis of Myosin Light Chain (LC) Isoforms

LC1emb, LC1f, LC1s, LC2s, and LC2f were present at 49 dg (data not shown), a stage when only primary myotubes were present (Fig. 3a–d). No LC3f was observed. The embryonic myosin light chain LC1emb was still present at 75 dg (Fig. 7a) and disappeared thereafter. Throughout the fetal development, the propor-

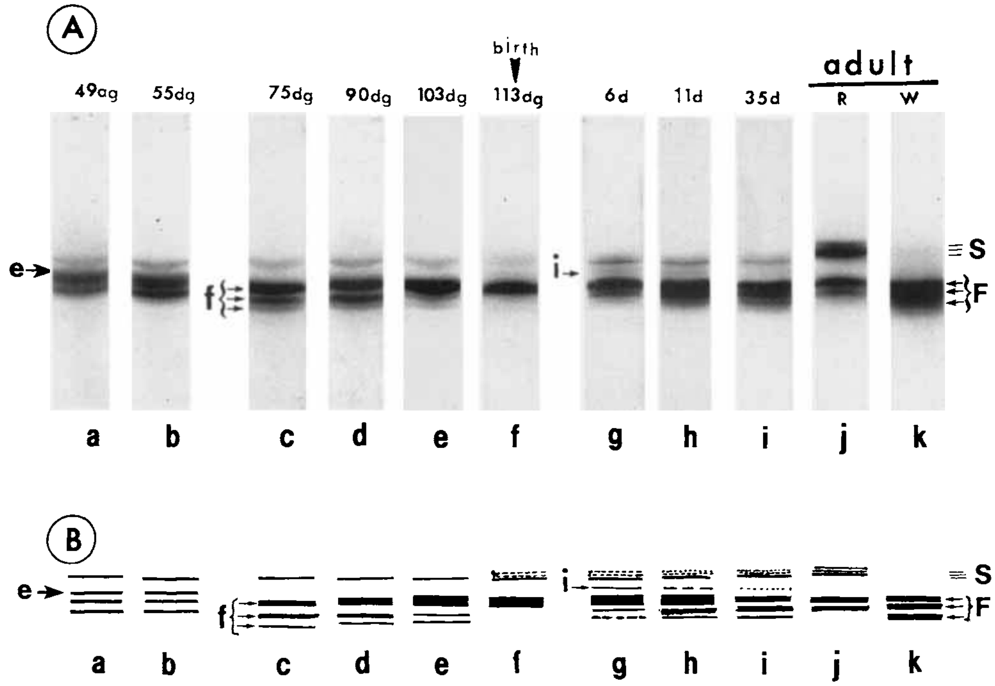


Fig. 2. Electrophoretic analysis of native myosin isozymes in developing semitendinosus (ST) muscle. The samples analyzed were whole ST from 49 (**a**), 55 (**b**), 75 (**c**), 90 (**d**), 103 (**e**), and 113 (**f**) days of gestation (dg), from 6- (**g**), 11- (**h**) and 35- (**i**) day-old piglets and from the deep medial red (R; **j**) and superficial white (W; **k**) portions of a 3-year-old adult. The proteins were stained with Coomassie blue R-250. The position of the bands corresponding to embryonic (e), fetal (f), adult slow (S), and adult fast (F) MHCs are indicated. An intermediate band (i) is detected during the early post-natal period.

tion of the fast isoforms LC1f and LC2f increased concomitantly with the formation of the second generation muscle fibers; this resulted in a relative decrease in the slow isoforms LC1s and LC2s. The proportion of the slow isoforms increased shortly after birth (Fig. 7c,d) as the second generation of type I fibers matured (Fig. 4i,m). The LC3f isoform appeared soon after birth (Fig. 7d) at the same time as the types IIA and IIB fibers began to attain a mature phenotype (Fig. 5). LC3f was highly expressed in the adult superficial portion of ST muscle (Fig. 7f). Two spots of variable intensity which correspond to the unphosphorylated (S2, F2) and phosphorylated (S2p, F2p) forms were frequently observed for LC2s and LC2f (Fig. 7e).

Immunocytochemical Analysis of Fetal MHC Expression

Up to 75 dg, all fibers in both portions homogenously expressed fetal MHC (Fig. 4b,d). Thereafter, primary fibers from the deep medial portion were the first to loose fetal MHC between 75 and 90 dg (Fig. 4b,f). Such a phenomenon could not be established in the superficial portion because primary and secondary fibers were indistinguishable from each other at 90 dg (Fig. 3p). Secondary fibers of both portions all stained for fetal MHC until at least 103 dg (data not shown). Thereafter, the reactivity to this antibody (Ab) decreased progressively and completely disappeared during the first two postnatal weeks (Fig. 4j,n,r, and l,p,t). A third generation of very small diameter fibers, labeled with the arabic numeral 3 on Figure 4r, appeared during the first two post-natal weeks. These fibers stained for fetal MHC and could not be identified on the other serial sections. However, the successive staining through 20 serial sections with this antibody confirmed they are muscle fibers.

Expression of Slow and Fast MHC in Primary Myotubes

At 49 dg, the two portions were apparent after staining with the anti-slow MHC Ab. In the deep medial region, primary myotubes laid close to one another and stained positively with slow MHC Ab (Fig. 3a). By contrast, primary myotubes were more dispersed and failed to react with this antibody in the superficial portion (Fig. 3c). From 49 dg to late gestation, slow MHC continued to be exclusively expressed in primary fibers from the deep medial portion (Fig. 3a,e,i,m, and c,g,k,o).

In both portions, primary myotubes reacted moderately and homogenously with the anti-fast antibody at 49 dg (Fig. 3b,d). At 55 dg, they stained very heterogenously in the deep medial portion (Fig. 3f), whereas they reacted strongly and uniformly in the superficial portion (Fig. 3h). At 75 dg, primary myotubes stained negatively with the anti-fast and positively with the

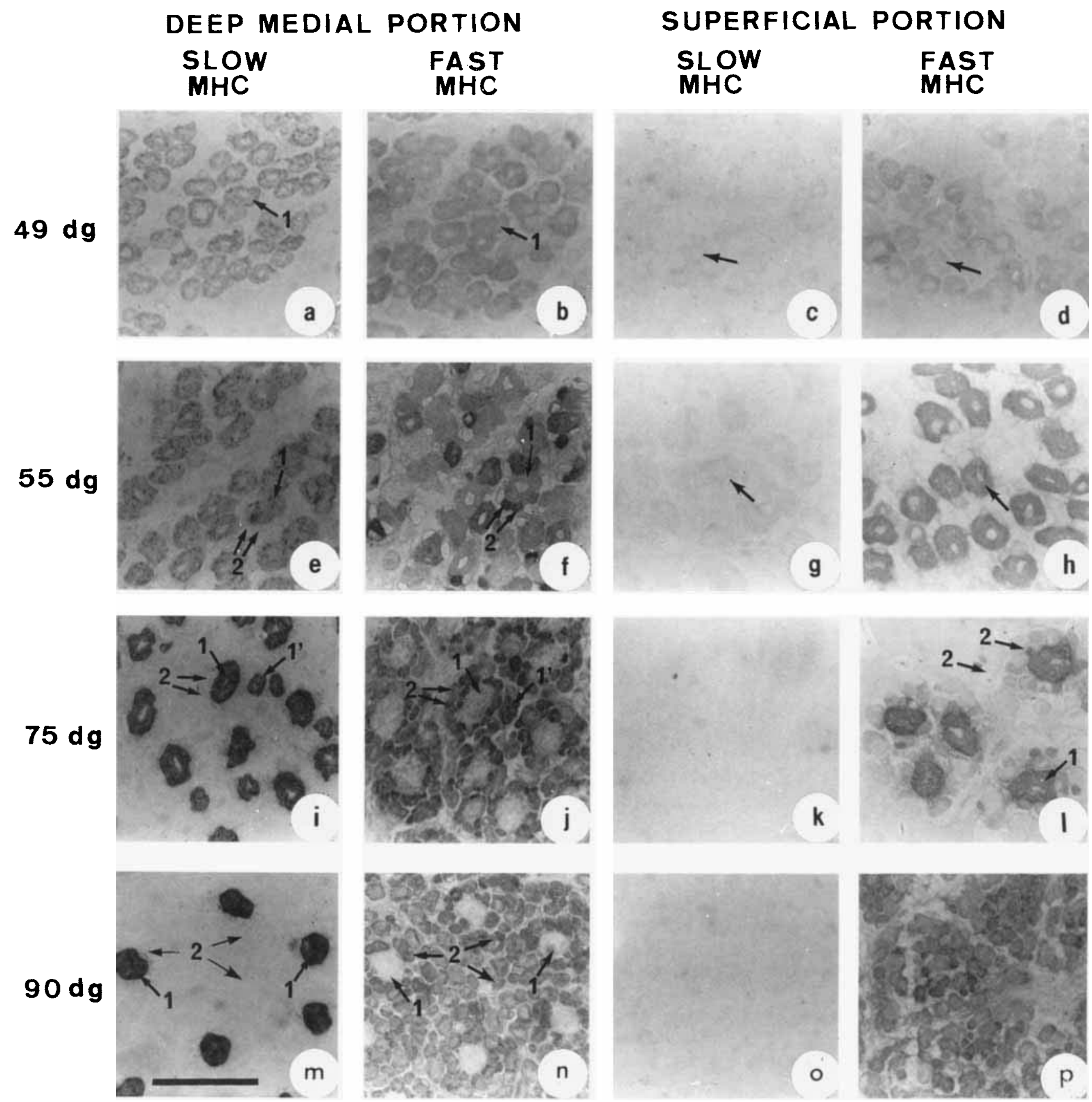


Fig. 3. Immunocytochemical analysis of the deep medial (**a,b,e,f,i, j,m,n**) and superficial (**c,d,g,h,k,l,o,p**) portions of semitendinosus muscle. Immunoperoxidase staining was carried out on serial sections at 49 (a,b,c,d), 55 (e,f,g,h), 75 (i,j,k,l), and 90 (m,n,o,p) days of gestation (dg). The antibodies used were against slow (a,b,e,f,i,j,m,n) and fast (c,d,g,h,k, l,o,p) myosin heavy chains (MHC). Arabic numerals on the photographs correspond to the primary (1) and secondary (2) generation fibers. Scale bars = 50 μm.

anti-slow in the deep medial portion (Fig. 3i,j). The opposite was observed in the superficial portion (Fig. 3k,l). The same pattern was observed at 90 dg (Fig. 3m,n,o,p). Subsequently, primary myotubes matured to type I fibers in the deep medial portion and type II fibers in the superficial portion. At 75 dg, a few myotubes of intermediate size strongly expressed both slow and fast MHC but were not surrounded by secondary fibers; they were named type 1′ to differentiate them from the classical type 1 primary myotubes (Fig. 3i,j). These type 1′ fibers may correspond to the isolated type I fibers sometimes encountered at the periphery of some islets in the adult muscle (see the type I fiber labeled with an arrow on Fig. 1A).

Expression of Slow and Fast MHC in Secondary Fibers

In both portions, secondary myotubes began to appear between 49 and 55 dg as small myotubes adjoining or in close proximity to the larger primary myotubes (Fig. 3f). They never expressed slow MHC until the end of gestation (Fig. 3a,e,i,m, and c,g,k,o). They stained positively with the anti-fast MHC as soon as they formed in the deep medial portion (Fig. 3f,j), and only from 75–90 dg onwards in the superficial portion (Fig. 3l,p). At 75 dg, secondary fibers still exhibited a smaller cross-sectional area than primary myotubes. At 90 dg, the number of secondary fibers surrounding each primary myotube had increased and in both portions of the muscle they reacted strongly with the anti-fast MHC Ab (Fig. 3n,p).

Slow MHC began to be expressed in a subpopulation of secondary fibers in the direct vicinity of primary fibers from late gestation onwards in the deep medial portion (Fig. 4i,m,q). Thus, the percentage of secondary fibers that expressed slow MHC gradually increased to reach a plateau of about 38% by 30 days after birth (Fig. 8). At the same time, these fibers progressively lost their reactivity with the anti-fast Ab (data not shown). This pattern of specialisation progressively gave rise to the typical disposition encountered in the pig, i.e., islets of type I fibers surrounded by type II fibers (Fig. 4i,m,q; Fig. 1A). A similar maturation of some secondary fibers to type I occurred in the superficial portion during the same time, but only 3% of the secondary fibers matured to become type I fibers. The location of type I fibers relative to the fast primary myotubes in the superficial portion could not be determined.

Expression of Alpha Cardiac MHC

In the deep medial portion, no fibers expressed α-cardiac MHC until 103 dg (data not shown). At 113 dg, the primary fibers expressed slow MHC but never α-cardiac MHC (Fig. 9a,b), whereas 16% of the secondary fibers, in the direct vicinity of the primary fibers, now expressed α-cardiac MHC (Fig. 9f). About 80% of these fibers also co-expressed slow MHC. Secondary fibers that expressed α-cardiac MHC were also the first of the secondary generation muscle fibers to express slow MHC. At 6 days, α-cardiac and slow MHC were present in the same secondary fibers representing 23% of the secondary fibers. During the subsequent maturation, the α-cardiac MHC was gradually eliminated and disappeared at about 80 days, whereas slow MHC continued to accumulate in other fibers so that 33% of the secondary fibers expressed slow MHC at 11 days (Fig. 8). Alpha-cardiac MHC was never expressed in primary myofibers nor in secondary fibers at the periphery of the islets.

In the superficial portion of the muscle, the nature of the phenomenon was similar to what happened in the deep medial portion but less pronounced. Fibers that started to express slow MHC during the first two postnatal weeks were also reactive with the anti-α-cardiac MHC Ab (Fig. 9k,l). At 11 days post-partum, only 3% of the fibers were reactive with the α-cardiac MHC Ab, as opposed to 23% in the deep medial portion at the same age. By analogy to what happened in the deep medial portion, these fibers are most likely secondary fibers.

DISCUSSION

Primary Myotubes Expressed Slow MHC in the Deep Medial Portion and Fast MHC in the Superficial Portion at 75 Days of Gestation

The absence of expression of slow MHC in primary myotubes of the superficial portion at 49 dg differed from previous results obtained in the fetal rat (Narusawa et al., 1987; Harris et al., 1989) where it was reported that there was a homogeneous expression of slow myosin in all primary myotubes in primordia of the future white and red portions of the anterior tibialis muscle at 16 days of gestation and a subsequent elimination of this isoform from the primary myotubes of the future white portion at about 10 days postpartum. The possibility remains that such a phenomenon occurs before 49 dg in the pig. In the present experiment, the maturation of primary myotubes towards type I fibers in the mixed deep red portion and type II fibers in the superficial fast white portion of the same muscle is in agreement with previous results reported in the anterior tibialis and crural muscles of the rat (Naruzawa et al., 1987; Condon et al., 1990a). However, it cannot be concluded that primary myotubes always mature to type II fibers in future fast twitch muscles. Indeed, they have been shown to give rise to type I fibers in the extensor digitorum longus, a fast twitch muscle (4% type I fibers), in the rat (Rubinstein and Kelly, 1981). Whether primary myotubes matured to type IIA, IIB, or IIX fibers in the superficial portion of ST muscle of the pig remains to be determined. The fate of primary myotubes to become slow fibers in the adult has also been reported in the sheep tibialis cranialis muscle, a mixed red muscle (Maier et al., 1992).

The early strikingly different expressions of slow MHC in primary myotubes between both portions could be neurally regulated since primary myotubes have been shown to be functionally innervated within hours of their formation (Dennis et al., 1981). This is consistent with the fact that primary myotubes are innervated before 45 dg in the pig ST muscle (Beermann et al., 1978). However, other studies carried out in the chick and rat have demonstrated that the initial generation and distribution of primary myotubes and fiber type pattern is independent of innervation or contractile activity (Harris, 1981; McLennan, 1983; Lyons et al., 1983; Miller et al., 1985; Crow and Stockdale, 1986; Condon et al., 1990b). In this case, the initial maturation of primary myotubes could be triggered by the presence of different classes of myoblasts with distinct

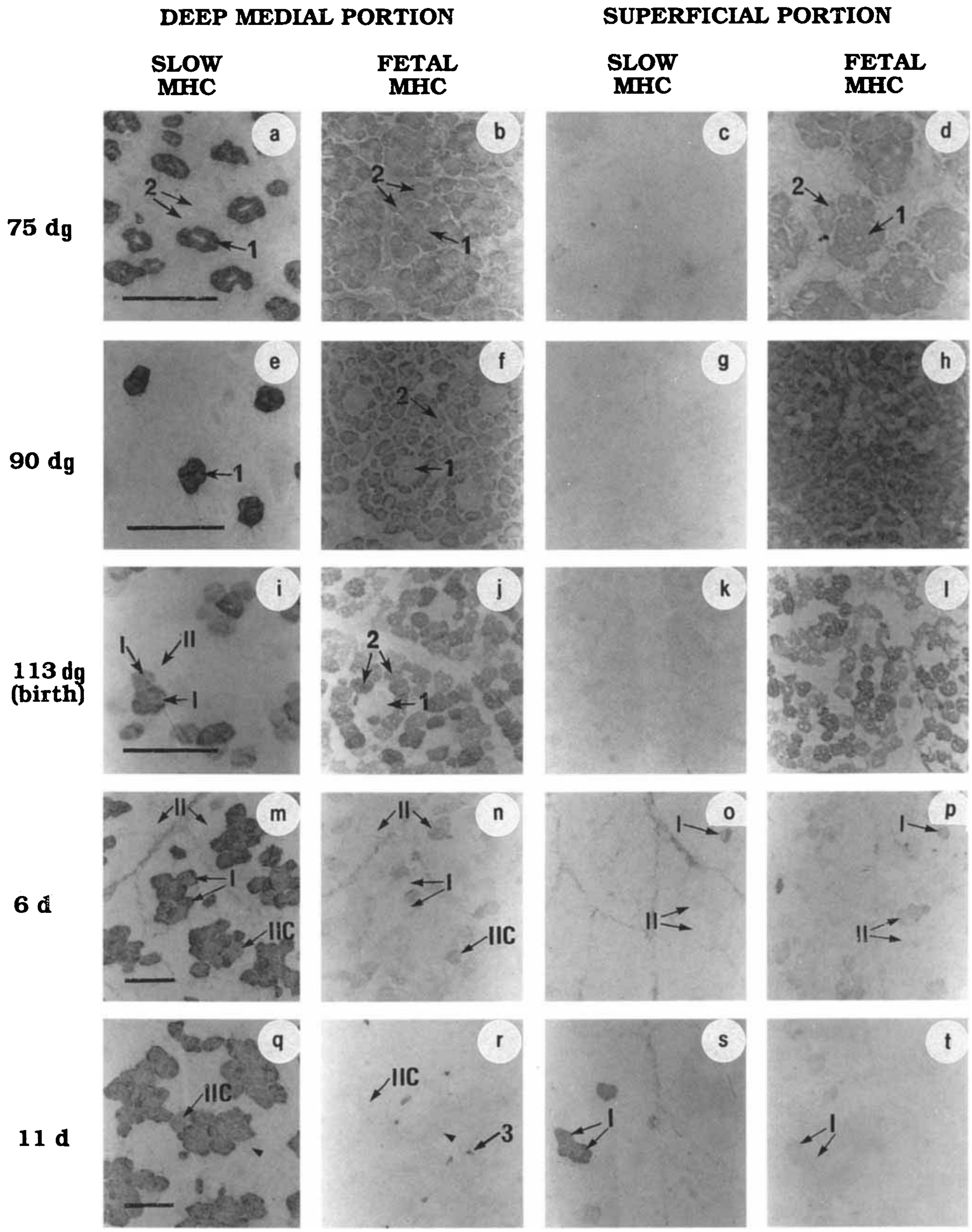

DEEP MEDIAL PORTION
SUPERFICIAL PORTION
SLOW MHC
FETAL MHC
SLOW MHC
FETAL MHC
75 dg
90 dg
113 dg (birth)
6 d
11 d

Fig. 4.

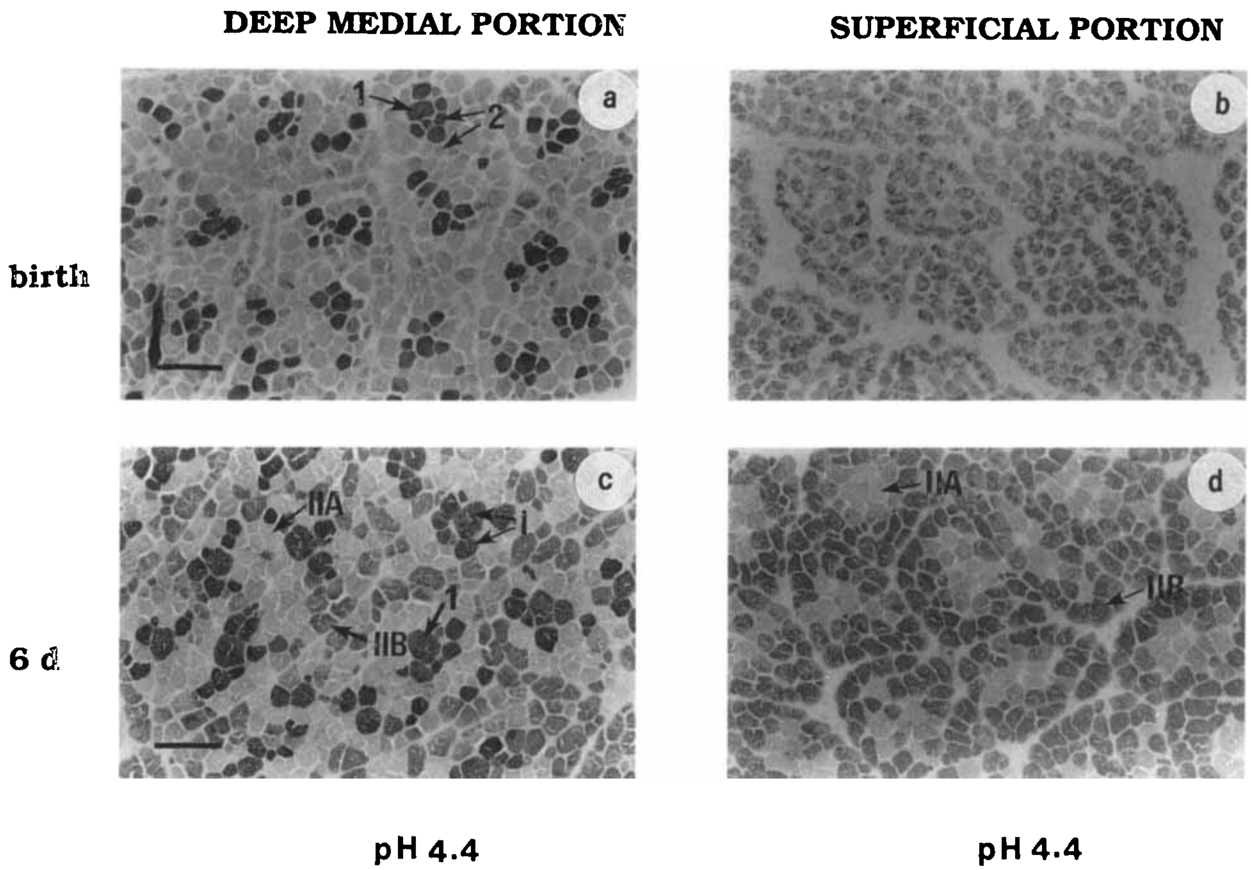


Fig. 5. ATPase activity after pre-incubation at pH 4.4 in the deep medial (**a,c**) and superficial (**b,d**) portions of pig semitendinosus muscle at birth (a,b) and at 6 days (c,d). Arabic and roman numerals on the photographs correspond to the generation (1, 2) and fiber type according to Brooke and Kaiser (1970) (I, IIA, IIB), respectively. Scale bar = 50 μm.

intrinsic programmes (Miller and Stockdale, 1986; Mouly et al., 1987) and/or by factors such as hormones, growth factors, and/or myogenic factors at the time the myotubes are formed. These different hypotheses could be tested in vitro by analysing cultures of myoblasts made from each portion.

Fetal and Adult Fast Isoforms of Native Myosin Comigrated

The overlapping of fetal (f1, f2, f3) and adult fast (F1, F2, F3) isoforms of native myosin (Fig. 2) has previously been observed in the cat (Hoh et al., 1988) but differs from other results in rabbit (Hoh and Yeoh, 1979), rat (Whalen et al., 1981; D'Albis et al., 1989; Laframboise et al., 1990), mouse (D'Albis et al., 1985),

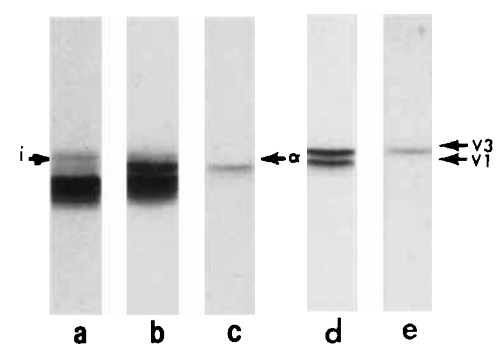


Fig. 6. Electrophoretic analysis of native myosin isozymes confirming the comigration of the intermediate band, called (i) in Figure 2, with the α-cardiac myosin. The samples analyzed were whole semitendinosus muscle extracts at 11 days post-partum (**a**), adult pig atrium (**c**), adult pig ventricle (**e**), a mixture of a and c (**b**), and of c and e (**d**). The atrium and ventricle are composed of αα and ββ MHC homodimers, respectively.

and man (Fitzsimons and Hoh, 1981b), in which fetal myosins migrated faster than adult fast native myosins. However, the disappearance of the isoforms f1 and f2 in late gestation and the loss of reactivity with the anti-fetal MHC Ab around birth (Fig. 4n,r, and p,t)

Fig. 4. Immunocytochemical analysis of the semitendinosus muscle. Immunoperoxidase staining was carried out on the deep medial (**a,b, e,f,i,j,m,n,q,r**) and superficial (**c,d,g,h,k,l,o,p,s,t**) portions of semitendinosus muscles at 75 (a,b,c,d), 90 (e,f,g,h), 113 (i,j,k,l) days of gestation (dg), and from 6- (m,n,o,p) and 11- (q,r,s,t) day-old piglets. The antibodies used were against slow (a,c,e,g,i,k,m,o,q,s) and fetal (b,d,f,h,j,l,n,p,r,t) myosin heavy chains (MHC). Arabic and roman numerals on the photographs correspond to the generation (1, 2) and fiber type according to Brooke and Kaiser (1970) (I, II), respectively. Scale bars = 50 μm.

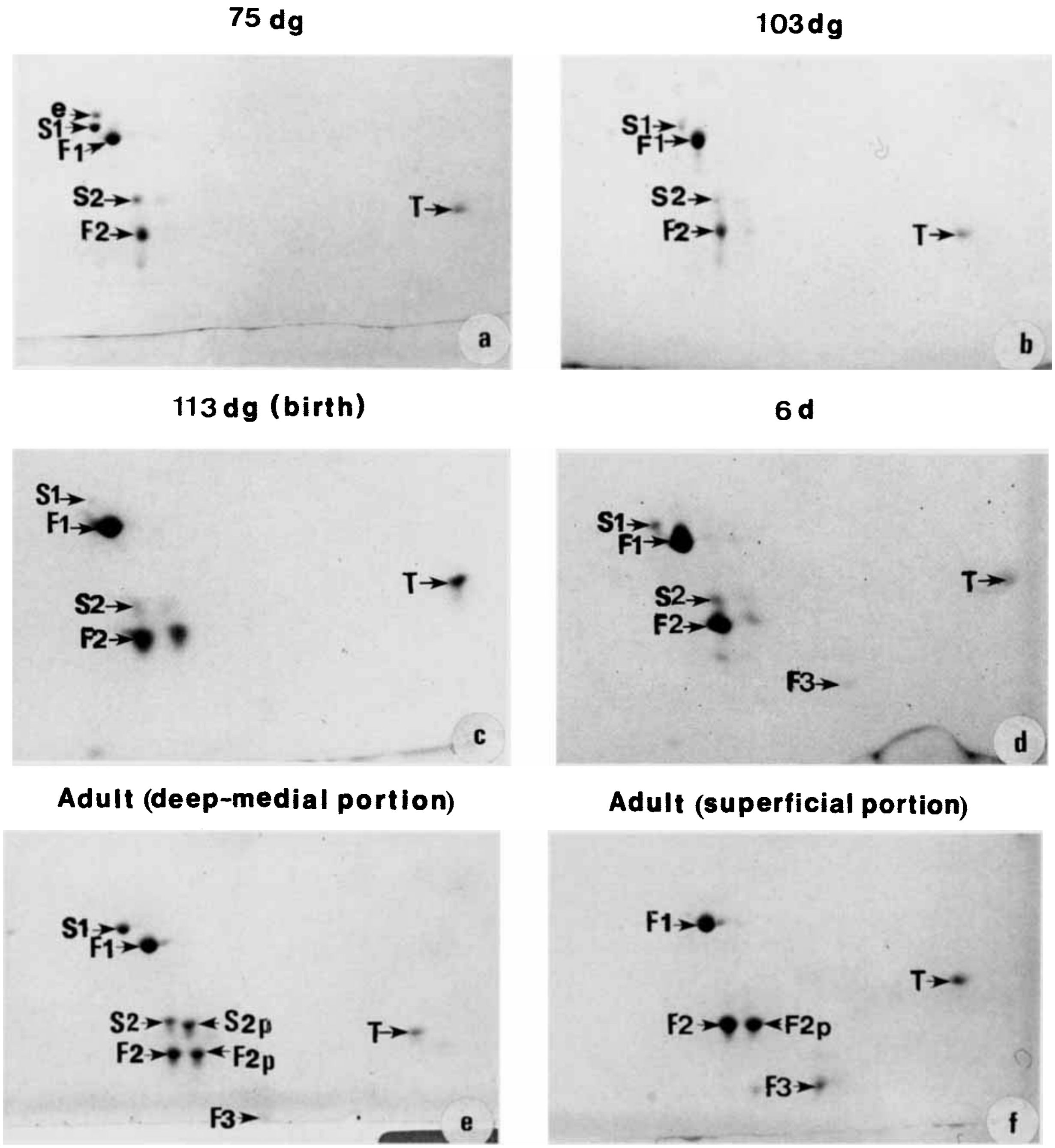


Fig. 7. The expression of myosin light chains during development of whole pig semitendinosus muscle. Two-dimensional gel electrophoresis of total muscle extracts at 75 (**a**), 103 (**b**), and 113 (**c**) days of gestation (dg), in 6- (**d**) day-old piglets and in the deep medial (**e**) and superficial (**f**) portions of a 3-year-old adult. The proteins were stained with Coomassie blue R-250. The various myofibrillar proteins are indicated as follows: embryonic myosin light chain (e), slow myosin light chains (S1 and S2), fast myosin light chains (F1, F2, F3), and troponin (T).

strongly suggest that these bands are fetal isoforms. The high increase in plasmatic thyroid hormone levels occurring at birth in the pig (Slebodzinski et al., 1981) could be involved in the mechanism leading to the disappearance of fetal isomyosin and accumulation of adult fast isoforms around birth, as suggested by previous experiments carried out in the rat (Gambke et al., 1983; Butler-Browne et al., 1984; D'Albis et al., 1990). One band migrating at the level of f3 and FM3 was present throughout the fetal and postnatal development. This band could be a mixture of fetal f3 and adult fast F3 isoforms during the fetal period, which could explain the positive staining of some fibers with the adult anti-fast MHC Ab during fetal development

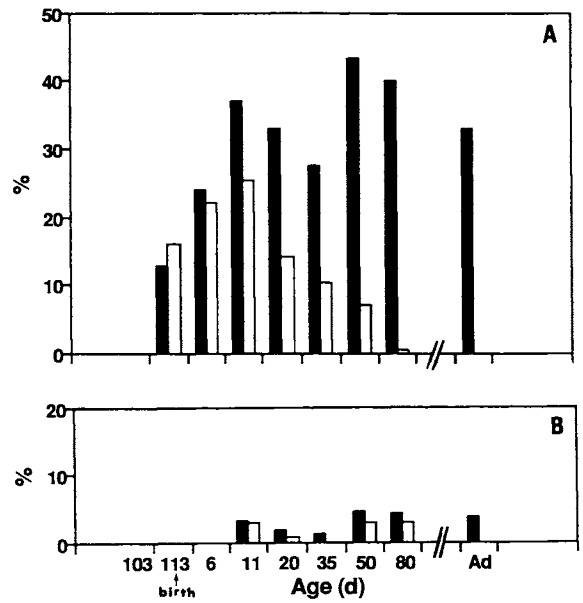


Fig. 8. Evolution in the proportion of secondary fibers containing slow myosin heavy chain (closed bars) and α-cardiac MHC (open bars) in the deep medial (**A**) and superficial (**B**) portions of the semitendinosus muscle at 103 and 113 days of gestation, in 6-, 11-, 20-, 35-, 50-, and 80-day-old piglets, and in 3-year-old adult (Ad).

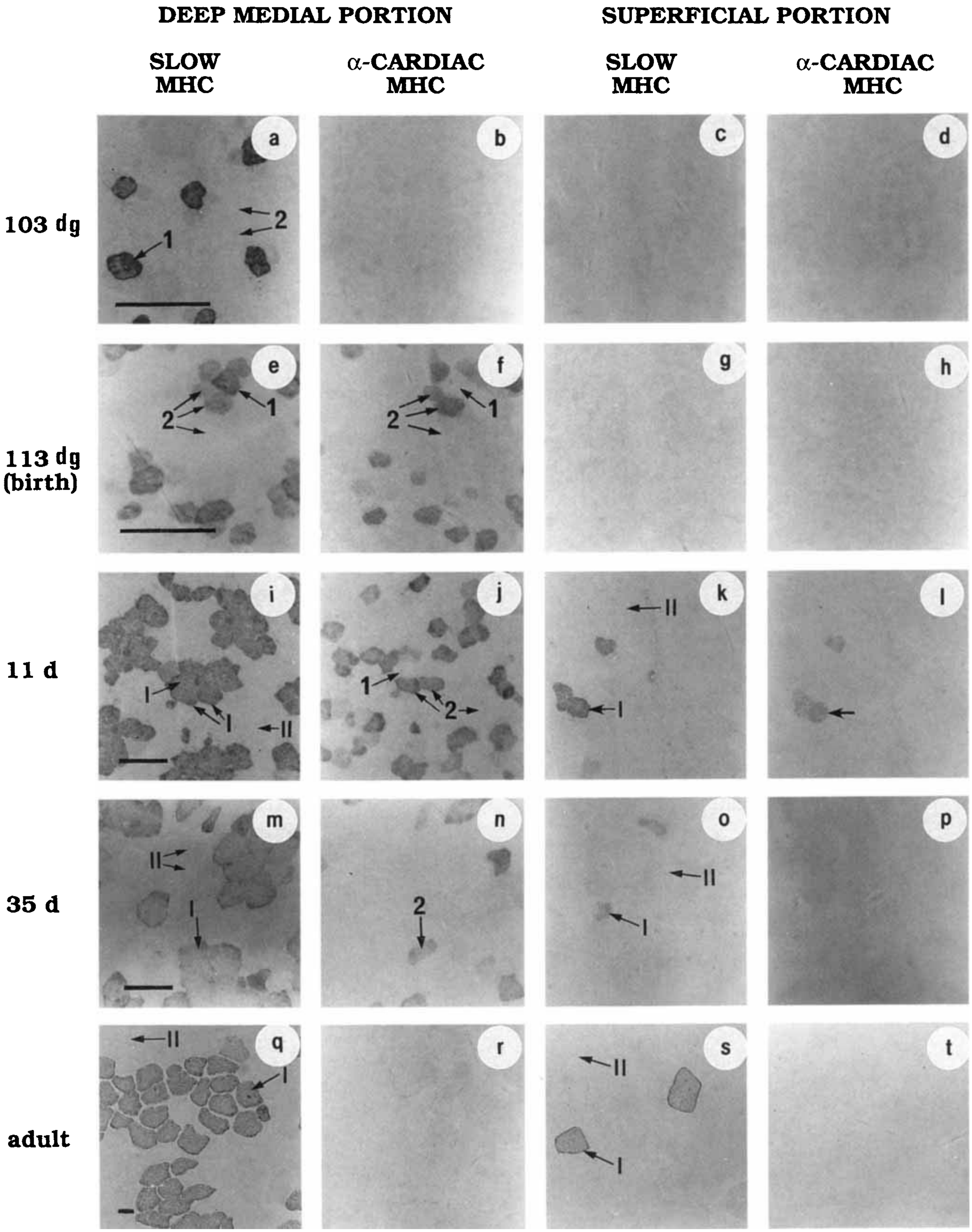
(Fig. 3b,f,j,n, and d,h,l,p). At 55 dg, the heterogenous staining of myofibers with the anti-fast MHC Ab (Fig. 3f), as opposed to the homogenous staining with the anti-embryonic and anti-fetal (data not shown), demonstrates, as in human (Ecob-Prince et al., 1989), that this anti-fast MHC Ab recognizes a MHC different from the actual embryonic and fetal MHCs. This MHC could be one of the adult fast MHC or another embryonic isoform, as yet unidentified in mammals, although known to exist in avian muscle (Dhoot, 1989). An early appearance of an adult fast MHC isoform has also been reported to occur from 13 weeks of gestation onwards in secondary fibers in human quadriceps muscle (Barbet et al., 1991) and in some secondary fibers located at the periphery of the rosettes at 76 days of gestation (birth occurs after 147 dg) in lamb tibialis cranialis (Maier et al., 1992). These last authors suggest that it is a IIB MHC; however, more work is needed to support this hypothesis, especially since the discovery of a third fast adult MHC, the type IIX (Schiaffino et al., 1989).

Secondary Fibers Expressed an Adult Fast MHC Isoform as Soon as They Formed in the Deep Medial Portion and Later on in the Superficial Portion

In the future fast white portion of ST, the fact that secondary fibers clearly reacted with the anti-fast Ab at only 90 dg (Fig. 3p), whereas they stained much earlier with the embryonic and fetal antibodies, suggests a sequential expression of MHC genes for embryonic, fetal, and adult fast myosin in these fibers, as previously reported in future fast muscles of the rat (Whalen et al., 1981; Mahdavi et al., 1986). However, this sequential expression did not seem to occur in the future red portion of ST muscle since secondary fibers stained positively with the anti-fast MHC Ab as soon as they formed (Fig. 3f,j).

A Subpopulation of Secondary Fibers Starts to Express Slow MHC From Late Gestation Onwards

In agreement with Rubinstein and Kelly (1981) in the rat, Barbet et al. (1990) in human, and Maier et al. (1992) in sheep, our results in the deep medial portion demonstrate that the induction of slow MHC synthesis is a biphasic phenomenon with an early expression in primary myotubes and a later expression in a subpopulation of secondary fibers which will mature to type I fibers. A striking difference can be observed in the pattern and timing of expression of the slow MHC in secondary fibers between the pig red ST and the sheep tibialis cranialis. In the pig, the de novo expression of slow MHC starts in late gestation and occurs in a subpopulation of secondary fibers in the direct vincinity of primary fibers, giving rise to the typical highly organized pattern of fiber type distribution we term rosettes, encountered in the deep medial portion of the adult pig ST (Fig. 1A). In the sheep tibialis cranialis muscle, this de novo expression occurs earlier (between 105 and 121 dg) and starts in secondary fibers which are not in the direct vicinity of primary fibers but regularly spaced within the muscle. Thereafter, additional transformation of secondary fibers to type I fibers changes the distribution of slow fibers to a clumped appearance with the disparition of the original organization in rosettes observed until 130 dg (Maier et al., 1992). The earlier transformation of secondary fibers to slow fibers in the sheep than in the pig is coherent with the well-known fact that the lamb is physiologically more mature at birth. The postnatal increase in the expression of slow myosin as demonstrated by immunocytochemistry was well correlated with the increase in proportion of type I fibers as revealed by enzyme histochemistry (data not shown) and the increase in the relative amounts of native myosin slow isoforms (Fig. 2, lanes f,g,h) and slow MLC (Fig. 7c,d). While accumulating slow MHC, these fibers progressively lost their reactivity to the antifast Ab. After 2 weeks of age, slow fibers issued from the second generation were morphologically and biochemically indistinguishable from those deriving from the primary myotubes, indicating a convergence of phenotypes with different developmental histories. The remaining secondary fibers expressed exclusively the adult fast MHC once the embryonic and fetal MHCs had been eliminated and matured to type IIA, IIB, or perhaps IIX. The distribution of fiber types encountered in the adult pig is also very different from the random distribution observed in the anterior tibialis of the rat (Narusawa et


DEEP MEDIAL PORTION
SUPERFICIAL PORTION
SLOW MHC
α-CARDIAC MHC
SLOW MHC
α-CARDIAC MHC
103 dg
113 dg (birth)
11 d
35 d
adult
a
b
c
d
e
f
g
h
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l
m
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o
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t
1
2
I
II

Fig. 9.

al., 1987) and the checkerboard distribution encountered in human muscle (Barbet et al., 1991).

The concomitant de novo perinatal expression of slow MHC and elimination of fast MHC in a subpopulation of secondary fibers in the direct vincinity of primary fibers could be neurally regulated. Indeed, it is generally assumed that muscle fibers undergo a neurally independent preprogrammed sequence of myosin transitions towards the fast phenotype (Embryonic → Neonatal → Fast) unless they receive a neural signal which will switch the programme towards the accumulation and maintenance of slow myosin, while inhibiting fast myosin synthesis (Salmons and Henriksson, 1981; Jolesz and Sreter, 1981; Butler-Browne et al., 1982; Condon et al., 1990b; Gambke et al., 1983; Butler-Browne and Whalen, 1984). The neural mechanisms leading to the formation of the islets of type I fibers are not understood, we only know that it is not due to the collateral branching of axons to innervate adjacent fibers (Beermann et al., 1978). Since the piglet can immediately walk and run after birth, the dramatic increase in contractile activity after birth may also be part of the mechanism.

Alpha Cardiac MHC Is Transitorily Expressed in a Subpopulation of Secondary Generation Muscle Fibers

Alpha cardiac MHC was so named because of its initial description in the heart. However, recent data have demonstrated that this isoform is also found in other skeletal muscles such as the masseter, retractor mandibulae, temporalis, diaphragm, and extraoccular muscles (Bredman et al., 1991; 1992a,b; D'Albis et al., 1991; Pedrosa-Domellöf et al., 1992). It is also present in some of the bag fibers of muscle spindles (Pedrosa et al., 1990; Pedrosa-Domellöf et al., 1992). However, this is the first study to demonstrate a transitory expression of an MHC isoform that is identical or closely related to α-cardiac MHC in a subpopulation of extrafusal fibers in a developing limb muscle. The negative staining of primary fibers with the anti-α-cardiac MHC mAb during the same time in the deep medial portion (Fig. 9f,j) demonstrates that this antibody reacts with an MHC different from the slow myosin isoform recognized by the anti-slow mAb. A similar transitory expression of α-cardiac MHC in the fibers starting to express slow MHC was observed in the superficial portion (Fig. 9k,l). By analogy to what happens in the deep medial portion, these fibers are likely to be secondary fibers. Interestingly, a transitory expression of α-cardiac MHC has also been reported to occur in the pig ventricle during the same period (Lompre et al., 1981). The transitorily expressed isoform we observed could also be a developmental MHC as suggested by the recent demonstration of the existence of three slow MHC isoforms in developing human and rat skeletal muscle (Hughes et al., 1993). Further research is still needed to determine the real nature of this isoform and we have undertaken to screen a cDNA library with this antibody to identify the mRNA coding for this isoform.

Since (1) thyroid hormones have been shown to directly increase the expression of α-cardiac MHC at the levels of mRNA and protein in the ventricular myocardium of rats and mice (Hoh et al., 1978; Lompre et al., 1984; Butler-Browne et al., 1987), (2) thyroid responsive elements have been identified in the rodent and human α-cardiac MHC gene promoter (Gustafson et al., 1987; Izumo and Mahdavi, 1988; Tsika et al., 1990), and (3) high levels of triiodothyronine (T3) and T4 are present during the early postnatal period in the pig (Slebodzinski et al., 1981), thyroid hormones could be the factor which triggers the concomitant transitory expression of α-cardiac MHC in both the heart (Lompre et al., 1981) and ST muscle (present study). In the first week following birth, TH returns to basal levels which are probably insufficient to maintain α-cardiac MHC expression. The selective action of TH on a subpopulation of second generation muscle fibers could be mediated either by a direct action of TH on responsive fibers or an indirect action of TH via the nerves. In both cases, neural input would most probably be the modulator of α-cardiac MHC expression.

A Third Generation of Small Diameter Fibers Forms Shortly After Birth

The presence of very small diameter fibers containing a developmental (fetal) MHC and forming during the early post-natal period (Fig. 4r) demonstrates the existence of a third generation of fibers and suggests a possible postnatal increase in the total number of fibers. This result contrasts with previous data reporting that the total number of myofibers is definitively established by 90 days of gestation in the pig (Thurley, 1972; Ashmore et al., 1973; Wigmore and Stickland, 1983). However, a third generation of fibers has also been reported in the longissimus of young piglets using an anti-fetal MHC antibody (Mascarello et al., 1992), and in human muscle at 16–17 weeks of gestation (Draeger et al., 1987) where they have been shown to contain fast and fetal MHC (Ecob-Prince et al., 1989). The presence of this third generation could be an important mechanism to explain muscle growth in large animals. However, further research is still needed to determine the origin and destiny of these tertiary fibers. We did not observe the appearance of a third generation of fibers filling in the space between the ro-

Fig. 9. Immunocytochemical analysis of the semitendinosus muscle. Immunoperoxidase staining was carried out on the deep medial (**a,b, e,f,i,j,m,n,q,r**) and superficial (**c,d,g,h,k,l,o,p,s,t**) portions of the semitendinosus muscle at 103 (a,b,c,d) and 113 (e,f,g,h) days of gestation (dg), in 11- (i,j,k,l) and 35- (m,n,o,p) day-old piglets and in a 3-year-old adult (q,r,s,t). The antibodies used were against slow (a,c,e,g,i,k,m,o,q,s) and α-cardiac (b,d,f,h,j,l,n,p,r,t) myosin heavy chains (MHC). Arabic and roman numerals on the photographs correspond to the fiber generation (1, 2) and fiber type according to Brooke and Kaiser (1970) (I, II), respectively. Scale bars = 50 μm.

settes, as suggested by Maier et al. (1992) in the lamb tibialis cranialis. Neither did our results allow us to determine the temporal and spacial pattern of formation of secondary fibers within the rosettes.

Three Adult Slow Isoforms of Native Myosin Are Expressed

Three slow isoforms of native myosin could be seen in the deep portion of ST muscle. These isoforms are the only ones present in the pure type I portion of the adult pig semimembranosus muscle where they are equally expressed (data not shown). Since preliminary observations did not reveal the presence of a second LC1s as reported in the rat (Whalen et al., 1978), whether this is due to the existence of different MHC isoforms or a different combination of MHC and MLCs still remains to be elucidated.

Concluding Remarks

The present work provides original data on the ontogenesis of the highly organized pattern of fiber type distribution in the deep medial and superficial portions of ST muscle in the pig (Fig. 10). In the deep medial portion, the central fiber of each islet of slow-twitch fibers originates from a primary myotube around which some secondary fibers in its direct vicinity matured to type I fibers between late gestation and the first postnatal weeks. The remaining secondary fibers matured to type IIA in the direct vicinity of these type I fibers and to type IIB at the periphery of the islets. In the superficial portion, primary myotubes and the majority of secondary fibers give rise to type II fibers. The remaining secondary fibers, representing about 3% of the secondary fibers, mature to become type I during the first 2 weeks following birth. In both portions of the muscle, a subpopulation of secondary fibers, the first ones to express slow MHC, also transitorily expressed an MHC that was identical or closely related to the α-cardiac MHC during the early postnatal period. This descripitive work is of great value to carry out future studies to identify these MHC and determine the mechanisms controling myofiber maturation during development. Due to the highly organized pattern of fiber type distribution in most of its muscles, the pig represents an interesting model whose comparison with other species can improve our understanding of muscle differentiation and maturation. Finally, from a practical point of view, these studies should contribute to identify the factors which are at the origin of the variability of muscle growth potential and meat quality in different breeds of pig.

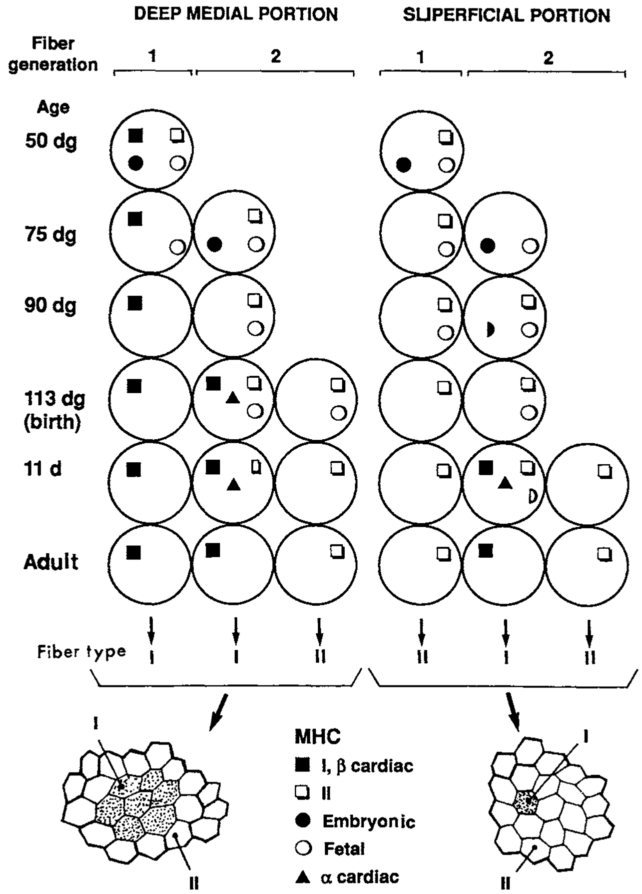

Fig. 10. A possible sequence of events leading to muscle fiber diversity in the deep-medial and superficial portions of the semitendinosus muscle of the pig. Embryonic and fetal myosin heavy chains (MHC) were successively eliminated in the primary and secondary fibers. Primary myotubes matured to type I in the deep-medial portion and to type II in the superficial portion. No expression of slow MHC occurred in the secondary fibers until the end of gestation. Around birth and during the first 2 weeks after birth, secondary fibers matured to become either type I or II in both portions. However, the conversion to type I was more important in the deep-medial (40%) than in the superficial (4%) portion. Around birth, a transitory expression of α-cardiac MHC occurred in a subpopulation of secondary fibers which were also the first ones to express slow MHC. These fibers were in the direct vicinity of primary myotubes in the deep medial portion, while their location could not be established in the superficial portion.

EXPERIMENTAL PROCEDURES

Animals

The semitendinosus (ST) muscle was removed from Large White pigs within 1 hr post-mortem. Samples were obtained from 49-, 55-, 75-, 90-, 103-, and 113-(birth) day-old fetuses, from 6-, 11-, 21-, 35-, 50-, and 80-day-old piglets, and from a 3-year-old pig. Artificial insemination allowed an accurate determination of fetal age. Muscle samples were frozen in isopentane cooled in liquid nitrogen and stored at −80°C until further analysis.

Biochemistry

Native myosin isoforms were separated on the basis of molecular weight using pyrophosphate gel electrophoresis in cylindrical tubes (6 × 0.5 cm) as previously described by Hoh et al. (1976), D'Albis et al. (1979), and Butler-Browne et al. (1990). Electrophoresis was carried out in a vertical gel apparatus Pharmacia GE 2/4 at 1–2°C (Pharmacia, Piscataway, NJ). Gels were

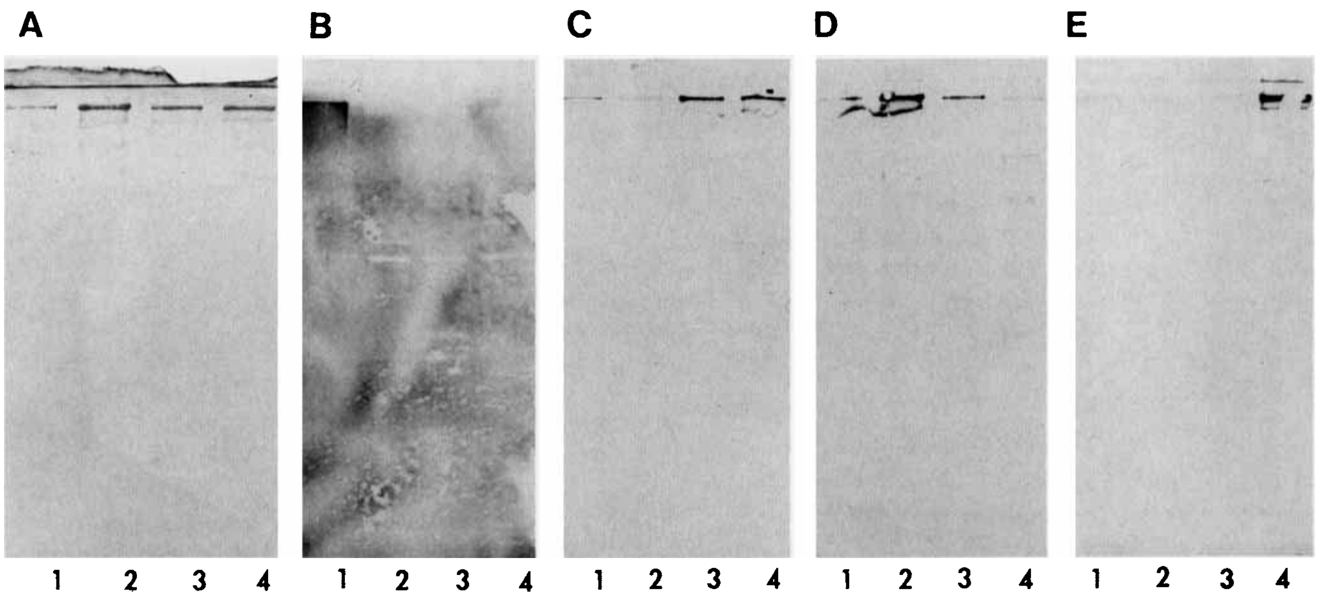

Fig. 11. Western blot assay of antibody binding to pig myosin heavy chains (MHC). Crude myosins were prepared by extracting muscle tissue in a high salt buffer and then precipitating once at a low ionic strength. After separation of the proteins on SDS-polyacrylamide gels, immunotransfers were prepared as described by Butler-Browne and Whalen (1984). In each panel the myosins are (**1**) 75 dg semitendinosus (ST), (**2**) superficial portion of adult ST, (**3**) deep red portion of semimembranosus, and (**4**) adult atrium. **A** shows a nitrocellulose paper stained with Coomassie blue. B through E show the reactions obtained using antibodies to (**B**) fetal, (**C**) adult slow, (**D**) adult fast, and (**E**) alpha-cardiac MHC.

stained with Coomassie blue R-250 and native myosin isoforms were identified according to Fitzsimons and Hoh (1981a). Two-dimensional gel electrophoresis of the myosin light chains (LC) was performed as previously described by O'Farrell (1975) and Butler-Browne et al. (1990). Briefly, isoelectric focusing in cylindrical gels was used to separate the proteins according to their charge in the first dimension, and SDS/polyacrylamide slab gels were run to separate the proteins according to their molecular weight in the second dimension. The gels were stained with Coomassie Brilliant Blue and the myosin light chains were identified according to Butler-Browne et al. (1990).

Enzyme Histochemistry

Ten micron serial transverse sections were cut in a cryostat at −20°C and air dried for histochemistry and immunocytochemistry. Three sections were stained for actomyosin adenosine triphosphatase (am-ATPase) activity after preincubation at pH 4.2, 4.4, and 10.4 (Guth and Samaha, 1970). Myofibers were classified as type I, IIA, IIB, and IIC (Brooke and Kaiser, 1970). The remaining serial sections were processed for immunocytochemistry.

Immunocytochemistry

Immunocytochemistry was carried out on 10 μm serial frozen sections using monoclonal antibodies (mAbs) against myosin heavy chains (MHC). The mAbs against human slow and fast MHC have been characterized by Ecob-Prince et al. (1989) and can be purchased from Novocastra (Newcastle, England). The anti-fast MHC mAb does not distinguish IIA from IIB MHC (Ecob-Prince et al., 1989). Monoclonal antibody F88 12F8 was raised against adult human cardiac myosin and has been shown to specifically recognize α-cardiac MHC in rat, cat, and man (Dechesne et al., 1985, 1987; Léger et al., 1985; Pons et al., 1986; Pedrosa-Domellöf et al., 1992). Antibody F88 4C10 was prepared against fetal bovine myosin and has been shown to be specific for fetal MHC (Robelin et al., 1993). The antibody reactivities with pig myosins are shown in Figure 11. Each panel is a nitrocellulose sheet containing proteins transferred from the SDS-PAGE of crude myosin preparations each containing predominantly one of the four myosin isozymes. Figure 11A has been stained with Coomassie blue to demonstrate all the proteins present on the transfer. Each of the antibodies shows considerable specificity of reaction with the four MHC. For example, anti-fetal Ab reacts only with fetal myosin (Fig. 11B, lane 1). The anti-slow reacts strongly with the deep red portion of semimembranosus muscle (Fig. 11C, lane 3) and negatively with the superficial portion of the adult ST (Fig. 11C, lane 2). The anti-fast mainly reacts with the superficial portion of the adult ST (Fig. 11D, lane 2), whereas the anti-alpha cardiac only reacts with the atrium (Fig. 11E, lane 4). The mAbs F88 12F8 and F88 4C10 were a generous gift from Drs. J. Léger and F. Pons (INSERM, U300, Montpellier, France). The sections were incubated overnight in a humid, sealed chamber at 4°C and specific antibody binding was revealed by the avidin biotin peroxidase technique (Vectastain ABC kit, Vector Laboratories, Burlingame, CA). Sections receiving only the second antibody showed no reaction and served as a control for background activity.

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REFERENCES

Ashmore, C.R., Addis, P.B., and Doerr, L. (1973) Development of muscle fibers in the fetal pig. J. Anim. Sci. 36:1088–1093.

Barbet, J.P., Thornell, L.-E., and Butler-Browne, G.S. (1991) Immunocytochemical characterisation of two generations of fibers during the development of the human quadriceps muscle. Mech. Dev. 35: 3–11.

Beermann, D.H., Cassens, R.G., and Hausman, G.J. (1978) A second look at fiber type differentiation in porcine skeletal muscle. J. Anim. Sci. 46:125–132.

Bredman, J.J., Wessels, A., Weijs, W.A., Korfage, J.A.M., Soffers, C.A.S., and Moorman, A.F.M. (1991) Demonstration of cardiac-specific myosin heavy chain in masticatory muscles of human and rabbit. Histochem. J. 23:160–170.

Bredman, J.J., Weijs, W.A., Korfage, H.A.M., Brugman, P., and Moorman, A.F.M. (1992a) Myosin heavy chain expression in rabbit masseter muscle during postnatal development. J. Anat. 180:263–274.

Bredman, J.J., Weijs, W.A., and Moorman, A.F.M. (1992b) Presence of cardiac α-myosin correlates with histochemical myosin ca2 + ATPase activity in rabbit masseter. Histochem. J. 24:260–265.

Brooke, M.H., and Kaiser, K.K. (1970) Muscle fiber types: How many and what kind? Arch. Neurol. 23:369–379.

Butler-Browne, G.S., and Whalen, R.G. (1984) Myosin isozyme transitions occurring during the postnatal development of the rat soleus muscle. Dev. Biol. 102:324–334.

Butler-Browne, G., Bugaisky, L., Cuenoud, S., Schwartz, K., and Whalen, R. (1982) Denervation of newborn rat muscles does not block the appearance of adult fast myosin heavy chain. Nature 299: 830–833.

Butler-Browne, G.S., Herlicoviez, D., and Whalen, R.G. (1984) Effects of hypothyroidism on myosin isozyme transitions in developing rat muscle. FEBS Lett. 166:71–75.

Butler-Browne, G.S., Prulière, G., Cambon, N., and Whalen, R.G. (1987) Influence of dwarf mouse mutation on skeletal and cardiac myosin isoforms. J. Biol. Chem. 262:15188–15193.

Butler-Browne, G.S., Barbet, J.P., and Thornell, L.E. (1990) Myosin heavy and light chain expression during human skeletal muscle development and precocious muscle maturation induced by thyroid hormone. Anat. Embryol. 181:513–522.

Condon, K., Silberstein, L., Blau, H., and Thompson, W.J. (1990a) Development of muscle fiber types in the prenatal rat hindlimb. Dev. Biol. 138:256–274.

Condon, K., Silberstein, L., Blau, H.M., and Thompson, W.J. (1990b) Differentiation of fiber types in aneural musculature of the prenatal rat hindlimb. Dev. Biol. 138:275–295.

Crow, M.T., and Stockdale, F.A. (1986) Myosin expression and specialization among earliest muscle fibers of the developing avian limb. Dev. Biol. 113:238–254.

D'Albis, A., Pantaloni, C., and Bechet, J.J. (1979) An electrophoretic study of native myosin isozymes and of their subunit content. Eur. J. Biochem. 99:261–272.

D'Albis, A., Janmot, C., and Bechet, J.J. (1985) Myosin switches in skeletal muscle development of an urodelan amphibian, *Pleurodeles walti*. Comparison with a mammalian, *Mus musculus*. Biochem. Biophys. Res. Commun. 128:94–100.

D'Albis, A., Couteaux, R., Janmot, C., and Roulet A. (1989) Specific programs of myosin expression in the postnatal development of rat muscles. Eur. J. Biochem. 183:583–590.

D'Albis, A., Chanoine, C., Janmot, C., Mira, J.C., and Couteaux, R. (1990) Muscle-specific response to thyroid hormone of myosin isoform transitions during rat postnatal development. Eur. J. Biochem. 193:155–161.

D'Albis, A., Janmot, C., Mira, J.-C. and Couteaux, R. (1991) Characterization of a ventricular V1 myosin isoform in rabbit masticatory muscles. Development and neural regulation. BAM 1:23–34.

Davies, A.S. (1972) Postnatal changes in the histochemical fibre types of porcine skeletal muscle. J. Anat. 113:213–240.

Dechesne, C.A., Léger, J.O.C., Bouvagnet, P., Claviez, M., and Léger, J.J. (1985) Fractionation and characterization of two molecular variants of myosin from adult human atrium. J. Mol. Cell. Cardiol. 17:753–757.

Dechesne, C.A., Léger, J.O.C., and Léger, J.J. (1987) Distribution of α- and β-myosin heavy chains in the ventricular fibers of the postnatal developing rat. Dev. Biol. 123:169–178.

Dennis, M.J., Ziskind-Conhaim, L., and Harris, A.J. (1981) Development of neuromuscular junctions in rat embryos. Dev. Biol. 81:266–279.

Dhoot, G.K. (1989) Evidence for the presence of a distinct embryonic isoform of myosin heavy chain in chicken skeletal muscle. Differentiation 40:176–183.

Draeger, A., Weeds, A.G., and Fitzsimons, R.B. (1987) Primary, secondary and tertiary myotubes in developing skeletal muscle: A new approach to the analysis of human myogenesis. J. Neurol. Sci. 81: 19–43.

Ecob-Prince, M., Hill, M., and Brown, W. (1989) Immunocytochemical demonstration of myosin heavy chain expression in human muscle. J. Neurol. Sci. 91:71–78.

Fitzsimons, R.B., and Hoh, J.F.Y. (1981a) Isomyosins in human type 1 and type 2 skeletal muscle fibers. Biochem. J. 193:229–233.

Fitzsimons, R.B., and Hoh, J.F.Y. (1981b) Embryonic and foetal myosins in human skeletal muscle. J. Neurol. Sci. 52:367–384.

Gambke, B., Lyons, G.E., Haselgrove, J., Kelly, A.M., and Rubinstein, N.A. (1983) Thyroidal and neural control of myosin transitions during development of rat fast and slow muscles. FEBS Lett. 156:335–339.

Gustafson, T.A., Markham, B.E., Bahl, J., and Morkin, E. (1987) Thyroid hormone regulates expression of a transfected α-MHC fusion gene in fetal heart cells. Proc. Natl. Acad. Sci. U.S.A. 84:3122–3126.

Guth, L., and Samaha, F.J. (1970) Research note: Procedure for histochemical demonstration of actomyosin ATPase. Exp. Neurol. 28: 365–367.

Handel, S.E., and Stickland, N.C. (1987) The growth and differentiation of porcine skeletal muscle fibre types and the influence of birthweight. J. Anat. 152:107–119.

Harris, A.J. (1981) Embryonic growth and innervation of rat skeletal muscles. I. Neural regulation of muscle fibre numbers. Philos. Trans. R. Soc. [Biol.] 293:257–277.

Harris, A.J., Fitzsimons, R.B., and McEwan, J.C. (1989) Neural control of the sequence of expression of myosin heavy chain isoforms in foetal mammalian muscles. Development 107:751–769.

Hoh, J.F.Y., and Yeoh, G.P.S. (1979) Rabbit skeletal myosin isoenzymes from foetal, fast-twitch and slow-twitch muscles. Nature (London) 280:321–323.

Hoh, J.F.Y., McGrath P.A., and White, R.I. (1976) Electrophoretic analysis of multiple forms of myosin in fast-twitch and slow-twitch muscles of the chick. Biochem. J. 157:87–95.

Hoh, J.F.Y., McGrath, P.A., and Hale, P.T. (1978) Electrophoretic analysis of multiple forms of rat cardiac myosin: Effects of hypophysectomy and thyroxine replacement. J. Mol. Cell. Cardiol. 11:1053–1076.

Hoh, J.F.Y., Hughes, S., Hale, P.T., and Fitzsimons, R.B. (1988) Immunocytochemical and electrophoretic analyses of changes in myosin gene expression in cat limb fast and slow muscles during postnatal development. J. Muscle Res. Cell Motil. 9:30–47.

Hughes, S.M., Cho, M., Karsch-Mizrachi, I., Travis, M., Silberstein, L., Leinwand, L.A., and Blau, H.M. (1993) Three slow myosin heavy chains sequentially expressed in developing mammalian skeletal muscle. Dev. Biol. 158:183–199.
Izumo, S., and Mahdavi, V. (1988) Thyroid hormone receptor isoforms generated by alternative splicing differentially activate MHC gene transcription. Nature (London) 334:539–543.
Jolesz, F., and Sreter, F.A. (1981) Development, innervation and activity-induced changes in skeletal muscle. Ann. Rev. Physiol. 43: 531–552.
Laframboise, W.A., Daood, M.J., Guthrie, R.D., Butler-Browne, G.S., Whalen, R.G., and Ontell, M. (1990) Myosin isoforms in neonatal rat extensor digitorum longus, diaphragm, and soleus muscle. Am. J. Physiol. 259:L116–L122.
Lefaucheur, L., and Vigneron, P. (1986) Postnatal changes in some histochemical and enzymatic characteristics of three pig muscles. Meat Sci. 16:199–216.
Léger, J.O.C., Bouvagnet, P., Pau, B., Roncucci., R., and Léger, J.J. (1985) Levels of ventricular myosin fragments in human sera after myocardial infarction determined with monoclonal antibodies to myosin heavy chains. Eur. J. Clin. Invest. 15:422–429.
Lompre, A.M., Mercadier, J.J., Wisnewsky, C., Bouveret, P., Pantaloni, C., D'Albis, A., and Schwartz, K. (1981) Species and age dependent changes in the relative amounts of cardiac myosin isoenzymes in mammals. Dev. Biol. 84:286–290.
Lompre, A.M., Nadal-Ginard, B., and Mahdavi, V. (1984) Expression of the cardiac alpha and beta myosin heavy chain genes is developmentally and hormonally regulated. J. Biol. Chem. 259:6437–6446.
Lyons, G.E., Haselgrove J., Kelly, A.L., and Rubinstein, N.A. (1983) Myosin transitions in developing fast and slow muscles of the rat hindlimb. Differentiation 25:168–175.
Mahdavi, V., Strehler, E.E., Periasamy, M., Wieczorek, D., Izumo, S., Grund, S., Strehler, M.A., and Nadal-Ginard, B. (1986) Sarcomeric myosin heavy chain family: Organization and pattern of expression. In: "Molecular Biology of Muscle Development," Emerson, D., Fischman, D., Nadal-Ginard, B., and Siddigui, M.A. (eds). New York: Alan R. Liss, Inc., pp 345–362.
Maier, A., McEwan, J.C., Dodds, K.G., Fischman, D.A., Fitzsimons, R.B., and Harris, A.J. (1992) Myosin heavy chain composition of single fibres and their origin and distribution in developing fascicles of sheep tibialis cranialis muscles. J. Muscle Res. Cell Motil. 13:551–572.
Mascarello, F., Stecchini, M.L., Rowlerson, A., and Ballocchi, E. (1992) Tertiary myotubes in postnatal growing pig muscle detected by their myosin isoform composition. J. Anim. Sci. 70:1806–1813.
McLennan, I.S. (1983) Neural dependence and independence of myotube production in chicken hind limb muscles. Dev. Biol. 98:287–294.
Miller, J.B., and Stockdale, F.E. (1986) Developmental regulation of the multiple myogenic cell lineages of the avian embryo. J. Cell. Biol. 103:2197–2208.
Miller, J.B., Crow, M.T., and Stockdale, F.E. (1985) Slow and fast heavy chain content defines three types of myotubes in early muscle cell cultures. J. Cell Biol. 101:1643–1650.
Mouly, V., Toutant, M., and Fiszman, M.Y. (1987) Chick and quail limb bud myoblasts, isolated at different times during development, express stage-specific phenotypes when differentiated in culture. Cell. Differ. 20:17–25.
Narusawa, M., Fitzsimons, R.B., Izumo, S., Nadal-Ginard, B., Rubinstein, N.A., and Kelly, A.M. (1987) Slow myosin in developing rat skeletal muscle. J. Cell Biol. 104:447–459.
O'Farrell, P.H. (1975) High resolution two dimensional electrophoresis of proteins. J. Biol. Chem. 250:4007–4021.
Pedrosa, F., Soukup, T., and Thornell, L.-E. (1990) Expression of an alpha cardiac-like myosin heavy chain in muscle spindle fibres. Histochemistry 95:105–113.
Pedrosa-Domellöf, F., Erikson, P.O., Butler-Browne, G.S., and Thornell, L.E. (1992) Expression of alpha-cardiac myosin heavy chain in mammalian skeletal muscle. Experientia 48:491–494.
Pette, D., and Staron, R.S. (1990) Cellular and molecular diversities of mammalian skeletal muscle fibers. Rev. Physiol. Biochem. Pharmacol. 116:1–76.
Pons, F., Léger, J.O.C., Chevallay, M., Tomé, F.M.S., Fardeau, M., and Léger, J.J. (1986) Immunocytochemical analysis of myosin heavy chains in human fetal skeletal muscles. J. Neurol. Sci. 76: 151–163.
Robelin, J., Picard, B., Listrat, A., Jurie, C., Barboiron, C., Pons, F., and Geay, Y. (1993) Myosin expression in semitendinosus muscle during foetal development of cattle: immunocytochemical and electrophoretic analyses. Reprod. Nutr. Dev. 33:25–41.
Rubinstein, N.A., and Kelly, A.M. (1981) Development of muscle fiber specialization in the rat hindlimb. J. Cell Biol. 90:128–144.
Salmons, S., and Henriksson, J. (1981) The adaptive response of skeletal muscle to increased use. Muscle Nerve 4:94–105.
Schiaffino, S., Gorza, L., Sartore, S., Saggin, L., Ausoni, S., Vianello, M., Gundersen, K., and Lomo, T. (1989) Three myosin heavy chain isoforms in type 2 skeletal muscle fibers. J. Musc. Res. Cell Motil. 10:197–205.
Slebodzinski, A.B., Nowak, G., and Zamyslowska, H. (1981) Sequential observation of changes in thyroxine, triiodothyronine and reverse triiodothyronine during the postnatal adaptation of the pig. Biol. Neonate 39:191–199.
Suzuki, A., and Cassens, R.G. (1980) A histochemical study of myofiber types in muscle of the growing pig. J. Anim. Sci. 51:1449–1461.
Swatland, H.J., and Cassens, R.G. (1973) Prenatal development, histochemistry and innervation of porcine muscle. J. Anim. Sci. 36: 343–354.
Thurley, D.C. (1972) Increase in diameter of muscle fibers in the fetal pig. Br. Vet. J. 128:355–358.
Tsika, R.W., Bahl, J., Leinwand, L.A., and Morkin, E. (1990) Thyroid hormone regulates expression of transfected human α-MHC fusion gene in fetal rat heart cells. Proc. Natl. Acad. Sci. U.S.A. 87:379–383.
Whalen, R.G., Butler-Browne, G.S., and Gros, F. (1978) Identification of a novel form of myosin light chain present in embryonic muscle tissue and cultured muscle cells. J. Mol. Biol. 126:415–431.
Whalen, R.G., Sell, S.M., Butler-Browne, G.S., Schwartz, K., Bouveret, P., and Pinset-Härstrom, I. (1981) Three myosin heavy-chain isozymes appear sequentially in rat muscle development. Nature 292:805–809.
Wigmore, P.M.C., and Stickland, N.C. (1983) Muscle development in large and small pig fetuses. J. Anat. 137:235–245.